Movie Memory Test: Event Boundaries and Movie Memory

BRSM Report: Team Odomos

Archit Choudhary (2023114002) Bhavya Ahuja (2023111035) Hrishiraj Mitra (2023111037)

IIIT Hyderabad

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Background: Event Segmentation Theory

- Our brains **segment** continuous experience into discrete *events*
- Transitions = **event boundaries** (Zacks et al., 2007)
- Boundaries trigger an update of the internal *event model*
- Memory at event boundaries is encoded **differently** (Swallow et al., 2009)

Two types of boundaries:

- **Natural boundaries (NB):** Smooth, organic transitions
- **Abrupt boundaries (AB):** Sudden, artificially imposed transitions (hard cuts)

Task Paradigm

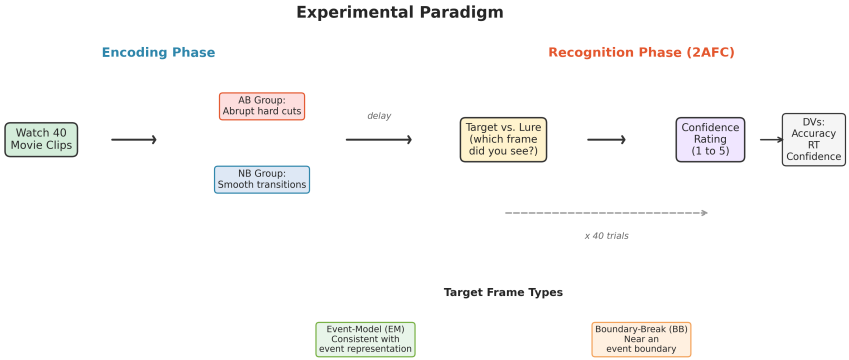


Figure 1: Experimental paradigm: encoding phase (AB vs. NB movie clips), followed by 2AFC recognition test with confidence ratings.

Design, Participants, and Demographics

- **2 $\times$ 2 mixed design:**
  - Boundary Type (AB vs. NB) – **between-subjects**
  - Target Frame Type (EM vs. BB) – **within-subjects**
- **170 participants** (81 AB, 89 NB); 1 excluded for missing data
- **Demographics:** Age $M = 22.2$ ($SD = 2.0$); 123 M, 47 F; 94 normal vision, 74 corrected
- Missing demographics (22/185) imputed: age $\rightarrow$ **median**, categoricals $\rightarrow$ **mode**
- **DVs:** Recognition accuracy, Response time, Confidence (1–5)

Six Hypotheses

	Hypothesis	Prediction
H1	Boundary Type $\rightarrow$ Accuracy	NB $>$ AB
H2	Boundary Advantage	BB $>$ EM (Radvansky & Zacks, 2017)
H3	Interaction $\rightarrow$ Accuracy	Condition $\times$ Target significant
H4	RT Interaction	AB: BB slower than EM; NB: no difference
H5	Signal Detection	NB higher d' than AB
H6	Demographics	Age, gender, vision moderate accuracy

Analytical Pipeline

- **171 PsychoPy CSV files** parsed in Python 3; **6,800 trials** final
- RT outliers removed: < 0.2 s or > 60 s (1 trial)

Extended pipeline for each DV:

1. Descriptive statistics (means, SDs)
2. **Normality check** (Shapiro-Wilk per cell, Levene's test)
3. 2×2 mixed ANOVA (η_p^2; follow-ups with Cohen's d)
4. Non-parametric robustness checks if normality violated
5. **Signal Detection Theory** (d', criterion c; log-linear correction)
6. **Mixed-effects models** (crossed random effects: subjects $\times$ items)
7. **Demographic moderation** (ANCOVA, correlations)

H1, H2: Accuracy Results

H1 (NB > AB): Supported

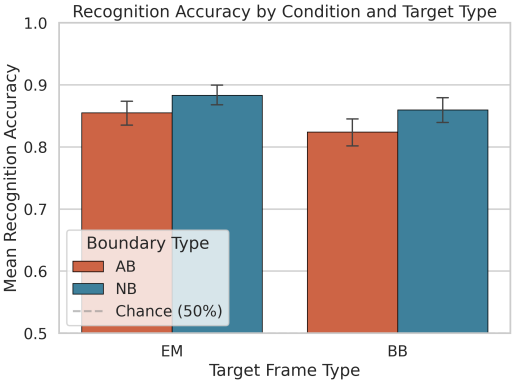
- $F(1, 168) = 7.25$, $p = .008$, $\eta_p^2 = .041$
- NB ($M = .871$) > AB ($M = .840$), $d = 0.41$

H2 (BB > EM): Not Supported

- $F(1, 168) = 11.44$, $p < .001$, $\eta_p^2 = .064$
- But EM ($M = .870$) > BB ($M = .843$)
- Direction **opposite** to boundary advantage prediction

H3 (Interaction): Not Supported

- $F(1, 168) = 0.21$, $p = .651$



H5: Signal Detection Theory

d' (Discriminability):

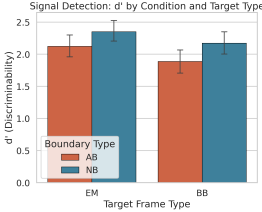
- NB d' = 2.26 > AB d' = 2.01
- $t = 2.55$, $p = .012$, $d = 0.39$
- **H5 supported**

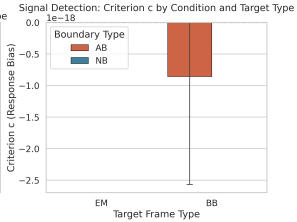
Criterion c (Response Bias):

- No group difference ($p = .296$)
- NB advantage is **genuine discriminability**

ANOVA on d':

- Condition: $F = 6.48$, $p = .012$
- Target: $F = 8.20$, $p = .005$





Mixed-Effects Models

Crossed random effects (subjects $\times$ items):

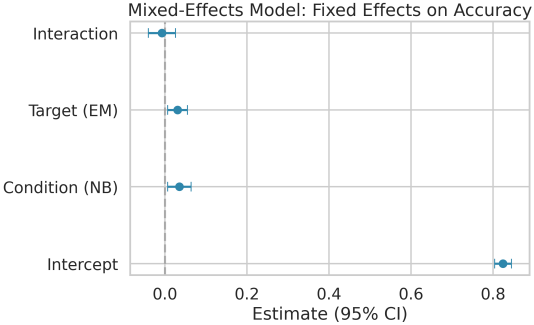
Accuracy:

- Condition: $b = .036$, $z = 2.95$, $p = .003$
- Target: $b = .031$, $z = 2.51$, $p = .012$
- Interaction: n.s. ($p = .669$)

Confidence:

- Condition: $b = .161$, $z = 6.89$, $p < .001$
- Target: $b = .091$, $z = 2.89$, $p = .004$

$\Rightarrow$ Results robust to item-level variability



H4: RT Interaction

Omnibus: $F(1, 168) = 3.36$, $p = .069$

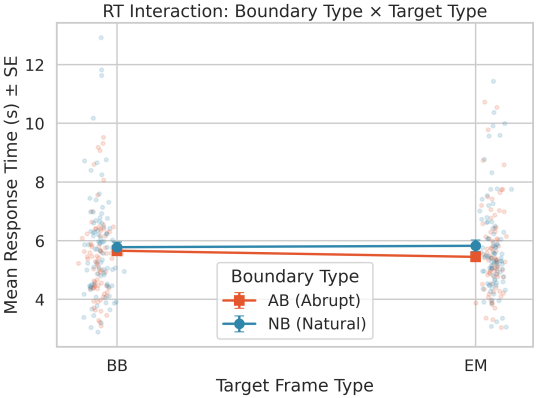
Simple effects:

- AB: EM (5.45 s) vs BB (5.66 s)
- $t = -2.56$, $p = .012$, $d = -0.15$
- NB: EM vs BB – n.s. ($p = .673$)

Bayesian analysis:

- $BF_{10} = 0.78$ (anecdotal for H0)
- Suggestive but inconclusive

$\Rightarrow$ **H4 partially supported**



Confidence Interaction and Dissociation

Confidence ANOVA:

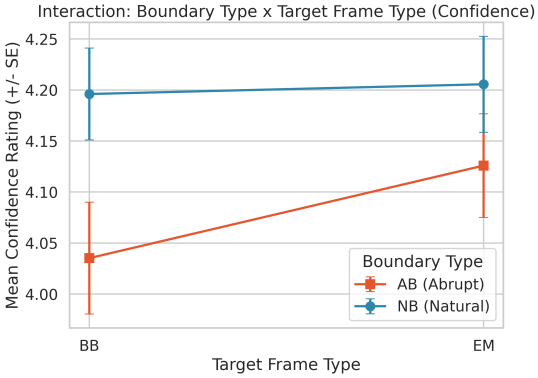
- Target Type: $p = .018$, $\eta_p^2 = .033$
- **Interaction: $p = .046$, $\eta_p^2 = .023$**

Simple effects:

- AB: EM $>$ BB ($p = .003$)
- NB: EM $\approx$ BB ($p = .733$)
- BB: NB $>$ AB ($p = .024$)

Key insight:

Abrupt boundaries impair **metacognitive certainty** for BB targets without proportionally reducing accuracy.



H6: Demographic Moderation

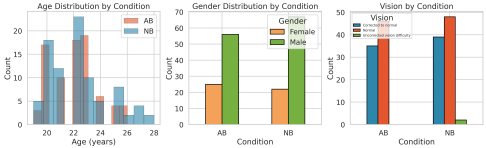
Age: Negative correlation with accuracy
$\rho = -0.203$, $p = .008$

Gender: Marginal female advantage
Females ($M = .875$) vs Males ($M = .849$), $p = .053$

Vision: No effect ($p = .749$)

ANCOVA: Condition effect **survives** covariate adjustment
$F = 8.38$, $p = .004$, $\eta_p^2 = .048$

$\Rightarrow$ **H6 partially supported** (age effect)



Summary of Results

Hypothesis	Prediction	Result	Support
H1 (Boundary $\rightarrow$ Acc)	NB $>$ AB	NB $>$ AB, $d =$ 0.41	**Yes**
H2 (Boundary advantage)	BB $>$ EM	EM $>$ BB (reversed)	**No**
H3 (Acc Interaction)	Significant	$p =$ .651	**No**
H4 (RT Interaction)	AB: BB slower	AB: $p =$ .012; NB: n.s.	**Partial**
H5 (SDT d')	NB $>$ AB	NB $>$ AB, $d =$ 0.39	**Yes**
H6 (Demographics)	Moderation	Age: $p =$ .008	**Partial**

All primary conclusions robust to non-parametric testing and mixed-effects modelling.

Conclusion and Future Directions

Key findings:

1. Natural boundaries facilitate encoding (accuracy *and* d')
2. Boundary advantage fails when boundaries are artificially disrupted
3. Abrupt cuts impair metacognitive certainty (confidence interaction)
4. Results robust across ANOVA, SDT, mixed-effects, and non-parametric tests

Future directions:

- Yes/no recognition paradigm for richer SDT analysis
- Model item-level predictors (boundary salience, lure similarity)
- Replicate RT interaction with larger sample
- Within-subjects boundary manipulation

Thank You

Team Odomos

Archit Choudhary | Bhavya Ahuja | Hrishiraj Mitra

Questions?

References

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